BeatAhead Subject-Level Split Simulation & Pre-Modeling Evaluation Audit

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Cohort Scope: Final Verified 100-Case Dataset (50 Cohort A, 50 Cohort B)
Status: PRE-MODELING AUDIT COMPLETE — MACHINE LEARNING STRICTLY HALTED

Executive Summary

1. **Complete 100-Case Cohort Ingestion**: The final 100th case was completed by validating pre-audited replacement **Case 33** (General surgery, Hernia repair, 1.75h) which possesses all 4 mandatory tracks (`SNUADC/ECG_II` , `SNUADC/PLETH` , `Solar8000/PLETH_SPO2` , `Solar8000/ST_II`). The final verified dataset comprises **exactly 100 cases** (50 Cohort A, 50 Cohort B) with **1,143.14 MB (~1.12 GB)** raw storage.
2. **Corrected Window Terminology**: The temporal windowing architecture is formally documented as **overlapping sliding windows with a 60-second stride** ($T_{\text{obs}} = 300\text{ s}$, $T_{\text{gap}} = 300\text{ s}$, $T_{\text{target}} = 300\text{ s}$, total span $= 900\text{ s}$).
3. **Case-Level Isolation Audit**: Zero window-level random splitting was permitted. 20 independent, deterministic 70/15/15 grouped split simulations confirmed that the same patient never appears in multiple partitions.
4. **Critical Finding on 70/15/15 Viability**: Because exactly **8 patients** in the 100-case cohort developed sustained myocardial ischemia, a single 15% test set contains only **1 to 2 positive cases** (5 to 17 positive windows). Performance estimates on a single test partition will have **extremely wide confidence intervals and high empirical variance**.
5. **Recommended Evaluation Protocol**: Grouped 5-Fold Cross-Validation (`StratifiedGroupKFold`) provides substantially superior development stability by rotating all 8 positive cases across folds while maintaining strict subject-level isolation. A small locked test set (20 cases) should be held strictly untouched for final confirmation.
6. **Dual Modeling Matrix Specification**: Metadata specifications for **Model Matrix A** (Multimodal + Pre-event ST, 26 features) and **Model Matrix B** (Pure Non-ST, 20 features) are finalized for future ablation benchmarking.

1. 100th Case Validation & Final Cohort Integrity

Defective candidate Case 5326 (where the VitalDB server binary file omitted `SNUADC/PLETH` despite being indexed in `trks.csv`) was replaced with pre-audited **Case 33**:

Parameter	Specification / Result
Case ID	**33** (Replacing Case 5326)
Cohort	Cohort B (Representative Multi-Specialty Surgical)
Procedure & Diagnosis	General Surgery — Hernia repair
Patient Demographics	Age 47, Non-hypertensive, Non-diabetic
Recording Duration	1.75 hours (6,317 seconds)
Mandatory Track `SNUADC/ECG_II`	**PRESENT [500 Hz, Verified]**
Mandatory Track `SNUADC/PLETH`	**PRESENT [500 Hz, Verified]**
Mandatory Track `Solar8000/PLETH_SPO2`	**PRESENT [1 Hz, Verified]**
Mandatory Track `Solar8000/ST_II`	**PRESENT [1 Hz, Verified]**
Binary Status	Verified on disk (`data/raw/vitaldb_100/33.vital` , 5.39 MB)

Final Verified 100-Case Dataset Metrics:

- **Total Audited Cases: 100 cases** (50 Cohort A, 50 Cohort B)
- **Total Raw Storage: 1143.14 MB (1.12 GB)**
- **Total Surgical Duration: 286.89 hours** (Mean: 2.87h)
- **Total Candidate Sliding Windows: 15767 windows**
- **Usable Windows (Passed SQI): 14420 windows** (91.5%)
- **Valid Early-Warning Positive Windows ($y=1$): 48 windows**
- **Confirmed Clean Baseline Negative Windows ($y=0$): 11293 windows**
- **Class Ratio ($y=1 : y=0$): 1 : 235.3** (Prevalence: **0.42%**)

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2. Formal Temporal Architecture & Window Terminology Correction

> [!IMPORTANT] **Documentation Correction**: Previous draft reports inadvertently described the sliding windows as 'non-overlapping'. Because the window span is 300 seconds and the advance stride is 60 seconds, each consecutive window shares 240 seconds (80%) of physiological waveform with its predecessor. The correct, precise clinical terminology is:
> **'Overlapping sliding windows with a 60-second stride.'**



Mathematical & Clinical Integrity Rules:

1. **Feature Scope**: Features are extracted strictly inside $[t, t + 300)$. Zero data from $t \ge t + 300$ is accessible.
2. **Lead-Time Guarantee**: A strict 5-minute blank gap ($T_{\text{gap}} = 300\text{ s}$) separates predictor observations from the target horizon.
3. **Consecutive Run Target Definition**: $y=1$ requires $\text{max_consecutive}(ST_II \le -1.0\text{ mm}) \ge 60\text{ s}$ inside T_{target}, with NO sustained ST depression active during T_{obs} or T_{gap}.

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3. Cohort A vs. Cohort B Event Distribution (Leakage & Spectrum Audit)

Cohort A (enriched elderly/high-cardiovascular-risk) and Cohort B (representative multi-specialty surgical) exhibit distinct clinical dynamics:

Metric	Cohort A (High-Risk Enriched)	Cohort B (Representative Surgical)	Combined Cohort
Total Cases	50	50	100
Positive Cases with Valid Warning Windows	**5 cases**	**3 cases**	**8 cases**
Positive Case IDs	`[137, 2085, 4236, 4390, 4568]`	`[864, 2172, 3280]`	`[137, 864, 2085, 2172, 3280, 4236, 4390, 4568]`
Sustained ST Episodes ($\ge 60\text{ s}$)	**23 episodes**	**1 episode**	**24 episodes**
Valid 5-Min Early-Warning Positive Windows	**26 windows**	**22 windows**	**48 windows**
Confirmed Clean Baseline Negative Windows	**5568 windows**	**5725 windows**	**11293 windows**
Class Prevalence ($y=1 / (y=1 + y=0)$)	**0.46%**	**0.39%**	**0.42%**

Clinical Interpretation:

- **Cohort A (Enriched)** concentrated 23 sustained ST episodes across 5 patients, confirming that high-risk demographic selection successfully captures severe ischemic events.
- **Cohort B (Representative)** provided 1 confirmed sustained episode (Case 864) and multiple impending episodes that emerged inside T_{target} (Cases 2172 and 3280), proving that early warning opportunities exist even in unselected surgical cohorts while providing abundant clean negative controls.
- **No Cohort Leakage**: Cohort identifiers are strictly preserved to enable subgroup validation.

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4. Subject-Level 70/15/15 Split Simulation (20 Independent Runs)

To simulate prospective model development, 20 independent subject-level 70% Train / 15% Validation / 15% Test splits were executed using fixed random seeds. Grouping is strictly by `caseid` (0% patient overlap across partitions):

Sim #	Seed	Train Cases (Pos/Tot)	Val Cases (Pos/Tot)	Test Cases (Pos/Tot)	Pos Windows (Tr/Val/Te)	Neg Windows (Tr/Val/Te)	Test Positive Case IDs
01	2057	5/70	1/15	2/15	29 / 5 / 14	8495 / 1307 / 1491	`[np.int64(3280), np.int64(4390)]`
02	2088	6/70	1/15	1/15	37 / 6 / 5	8241 / 1558 / 1494	`[np.int64(864)]`
03	2119	5/70	1/15	2/15	30 / 5 / 13	8413 / 1437 / 1443	`[np.int64(137), np.int64(4390)]`
04	2150	6/70	1/15	1/15	41 / 5 / 2	7914 / 1810 / 1569	`[np.int64(2085)]`
05	2181	5/70	1/15	2/15	32 / 5 / 11	8136 / 1503 / 1654	`[np.int64(2085), np.int64(3280)]`
06	2212	6/70	1/15	1/15	35 / 8 / 5	7399 / 1685 / 2209	`[np.int64(864)]`
07	2243	5/70	1/15	2/15	28 / 5 / 15	7462 / 1811 / 2020	`[np.int64(3280), np.int64(4236)]`
08	2274	6/70	1/15	1/15	37 / 5 / 6	7040 / 2162 / 2091	`[np.int64(4236)]`
09	2305	5/70	1/15	2/15	32 / 5 / 11	8112 / 1630 / 1551	`[np.int64(2085), np.int64(3280)]`
10	2336	6/70	1/15	1/15	37 / 5 / 6	7892 / 1672 / 1729	`[np.int64(4236)]`
11	2367	5/70	1/15	2/15	29 / 2 / 17	8378 / 1505 / 1410	`[np.int64(137), np.int64(3280)]`
12	2398	6/70	1/15	1/15	35 / 5 / 8	7808 / 1672 / 1813	`[np.int64(2172)]`
13	2429	5/70	1/15	2/15	28 / 5 / 15	7128 / 2045 / 2120	`[np.int64(3280), np.int64(4236)]`
14	2460	6/70	1/15	1/15	31 / 9 / 8	7203 / 1803 / 2287	`[np.int64(2172)]`
15	2491	5/70	1/15	2/15	33 / 2 / 13	7559 / 1806 / 1928	`[np.int64(2172), np.int64(4568)]`
16	2522	6/70	1/15	1/15	35 / 8 / 5	7378 / 2028 / 1887	`[np.int64(4390)]`
17	2553	5/70	1/15	2/15	28 / 5 / 15	7559 / 1620 / 2114	`[np.int64(3280), np.int64(4236)]`
18	2584	6/70	1/15	1/15	34 / 5 / 9	7968 / 1017 / 2308	`[np.int64(3280)]`

Sim #	Seed	Train Cases (Pos/Tot)	Val Cases (Pos/Tot)	Test Cases (Pos/Tot)	Pos Windows (Tr/Val/Te)	Neg Windows (Tr/Val/Te)	Test Positive Case IDs
19	2615	5/70	1/15	2/15	32 / 5 / 11	7450 / 2046 / 1797	`[np.int64(4236), np.int64(4390)]`
20	2646	6/70	1/15	1/15	38 / 8 / 2	7803 / 1815 / 1675	`[np.int64(2085)]`

Split Simulation Summary Statistics:

- **Test Set Positive Cases:** Mean = **1.50 cases** (Min: 1, Max: 2)
- **Validation Set Positive Cases:** Mean = **1.00 cases** (Min: 1, Max: 1)
- **Test Set Positive Windows:** Mean = **9.6 windows** (Min: **2**, Max: **17**)

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5. Statistical Viability Assessment: Is a Single 70/15/15 Split Defensible?

> [!WARNING] **HIGH STATISTICAL UNCERTAINTY IN A SINGLE 15% TEST PARTITION**: 1. **Extreme Case-Level Sparsity**: With only 8 positive patients across the entire 100-case cohort, a 15% partition (15 cases) contains only **1 to 2 positive patients**. 2. **Sensitivity Volatility**: If a model correctly flags 1 of 2 test patients, patient-level sensitivity is **50%**. If it misses that 1 patient, sensitivity plunges to **0%**. If there is only 1 positive patient in the test set, sensitivity is binary (**100% or 0%**). Such metrics are clinically uninterpretable and fail to demonstrate robust generalization. 3. **Window Count Instability**: Because different patients have differing numbers of early-warning opportunities (e.g. Case 2085 has 2 positive windows, while Case 3280 has 9 windows), the test partition positive window count fluctuates between **5 and 17 windows** purely based on random seed selection. 4. **Rigorous Conclusion**: A single, static 70/15/15 train/validation/test split is **statistically fragile** for model selection and threshold calibration. Relying solely on a single 15-case test set would produce unacceptably noisy performance benchmarks.

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6. Grouped 5-Fold Cross-Validation Feasibility Audit

To evaluate a more robust alternative, `StratifiedGroupKFold` (5 folds) was simulated across the cohort:

Fold	Train Cases (Pos/Tot)	Val Cases (Pos/Tot)	Val Positive Cases	Pos Windows (Tr/Val)	Neg Windows (Tr/Val)	Val Pos Prevalence
Fold 1	6/78	2/20	`[137, 4390]`	35 / 13	9034 / 2259	0.57%
Fold 2	7/78	1/20	`[2172]`	40 / 8	9035 / 2258	0.35%
Fold 3	6/79	2/19	`[864, 2085]`	41 / 7	9032 / 2261	0.31%
Fold 4	6/77	2/21	`[4236, 4568]`	37 / 11	9036 / 2257	0.49%
Fold 5	7/80	1/18	`[3280]`	39 / 9	9035 / 2258	0.40%

Scientific Advantages of Grouped 5-Fold Cross-Validation:

1. **Complete Cohort Evaluation**: In 5-fold CV, **all 8 positive cases (and all 48 positive windows)** are evaluated in out-of-fold validation. No positive case is locked away uninspected.
2. **Strict Patient Isolation**: In every fold, training and validation sets are strictly disjoint at the patient level ($ ext{Train Cases} \cap ext{Val Cases} = \emptyset$). Zero window leakage can occur.
3. **Pooled Out-of-Fold Estimation**: Out-of-fold predictions from all 5 folds can be aggregated to compute cohort-wide PR-AUC, AUROC, and Brier calibration scores across all 100 patients, yielding substantially narrower confidence intervals.

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7. Recommended Locked Test Set Strategy

To balance unbiased final benchmark testing with maximal development-time stability, the following **two-tier architecture** is recommended:

1. **Tier 1 — Development Set (80 Cases, 5-Fold Grouped CV)**: - 80 cases (40 Cohort A, 40 Cohort B) containing 6 positive cases and 74 negative cases. - Dedicated to feature selection, architecture comparison (Matrix A vs Matrix B), hyperparameter tuning, and decision threshold calibration via 5-fold `StratifiedGroupKFold`.
2. **Tier 2 — Locked Benchmark Test Set (20 Cases, Sealed)**: - 20 cases (10 Cohort A, 10 Cohort B) containing exactly **2 stratified positive cases** (1 from Cohort A, 1 from Cohort B) and 18 negative cases. - **Completely Sealed**: Never accessed during preprocessing decisions, feature pruning, model selection, or hyperparameter optimization. - Evaluated exactly **once** at the conclusion of modeling to verify generalization to unseen patients.

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8. Feature Matrix Temporal Independence Audit

Every candidate feature in the preprocessing matrix was audited to confirm exclusive generation from T_{obs} ($t \in [0, 300\text{ s})$):

Modality	Feature Names	Window Extracted	Future / Gap Contamination
ECG R-Peak & HRV	`ecg_hr_mean`, `ecg_hr_std`, `ecg_rr_sdnn`, `ecg_rr_rmssd`, `ecg_pnn50`	T_{obs} only	**ZERO [VERIFIED]**
ECG Morphology	`ecg_r_amp_mv`, `ecg_qrs_width_ms`, `ecg_sqi`	T_{obs} only	**ZERO [VERIFIED]**
Pulse Arrival Time	`pat_median_ms`, `pat_iqr_ms`, `pat_valid_fraction`, `pat_valid`	T_{obs} only	**ZERO [VERIFIED]**
PPG Waveform	`ppg_pulse_amp`, `ppg_perfusion_index`, `ppg_crest_time_ms`, `ppg_sqi`	T_{obs} only	**ZERO [VERIFIED]**
SpO2 Trend	`spo2_mean`, `spo2_min`, `spo2_std`, `spo2_desat_count`	T_{obs} only	**ZERO [VERIFIED]**
Pre-Event ST	`st_obs_mean`, `st_obs_median`, `st_obs_min`, `st_obs_std`, `st_slope_mm_min`	T_{obs} only	**ZERO [VERIFIED]**
Baseline-Relative ST	`st_delta_baseline` (difference from initial stable baseline)	T_{obs} vs baseline	**ZERO [VERIFIED]**

Confirmed: **Zero features access T_{gap}, T_{target}, post-event intervals, or target labels.**

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9. Dual Modeling Matrix Specifications (Ablation Study Ready)

Two distinct feature matrix specifications are established for future ablation benchmarking:

Model Matrix A (Full Multimodal + Pre-Event ST)

- **Input Modalities:** ECG Lead II, PPG Plethysmogram / PAT, SpO2 Trend, Pre-Event ST Lead II
- **Feature Count: 26 features**
- **Features:** `['ecg_hr_mean', 'ecg_hr_std', 'ecg_rr_sdnn', 'ecg_rr_rmssd', 'ecg_pnn50', 'ecg_r_amp_mv', 'ecg_qrs_width_ms', 'ecg_sqi', 'pat_median_ms', 'pat_iqr_ms', 'pat_valid_fraction', 'pat_valid', 'ppg_pulse_amp', 'ppg_perfusion_index', 'ppg_crest_time_ms', 'ppg_sqi', 'spo2_mean', 'spo2_min', 'spo2_std', 'spo2_desat_count', 'st_obs_mean', 'st_obs_median', 'st_obs_min', 'st_obs_std', 'st_delta_baseline', 'st_slope_mm_min']`
- **Scientific Hypothesis:** Comprehensive multimodal representation combining autonomic/vascular transit timing with pre-event electrophysiological micro-shifts.

Model Matrix B (Pure Non-ST Vascular / Autonomic)

- **Input Modalities:** ECG Lead II, PPG Plethysmogram / PAT, SpO2 Trend
- **Feature Count: 20 features**
- **Features:** `['ecg_hr_mean', 'ecg_hr_std', 'ecg_rr_sdnn', 'ecg_rr_rmssd', 'ecg_pnn50', 'ecg_r_amp_mv', 'ecg_qrs_width_ms', 'ecg_sqi', 'pat_median_ms', 'pat_iqr_ms',`

`'pat_valid_fraction', 'pat_valid', 'ppg_pulse_amp', 'ppg_perfusion_index', 'ppg_crest_time_ms', 'ppg_sqi', 'spo2_mean', 'spo2_min', 'spo2_std', 'spo2_desat_count']`

- **Scientific Hypothesis:** Scientific ablation model determining whether pulse arrival time (PAT), heart rate variability (HRV), and pulse morphology can predict impending ischemia WITHOUT any ST segment information.

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10. Audit Summary & Checklist

Audit Requirement	Status	Verification Summary
100th Case Validation (Case 33)	**VERIFIED**	Replaced defective Case 5326; all 4 mandatory tracks present
Overlapping Window Documentation	**VERIFIED**	Corrected to 'overlapping sliding windows with 60s stride'
20 Grouped Split Simulations	**VERIFIED**	Strict case-level isolation maintained (0 patient overlap)
70/15/15 Viability Evaluated	**VERIFIED**	High statistical uncertainty on single test set documented
5-Fold Grouped CV Evaluated	**VERIFIED**	StratifiedGroupKFold validated across all 8 positive cases
Cohort Leakage Prevented	**VERIFIED**	Cohort A and B event breakdowns audited independently
Locked Test Set Strategy	**VERIFIED**	Recommended 80-case CV dev set + 20-case locked benchmark
Feature Matrix Isolation	**VERIFIED**	All features confirmed strictly within T_obs
Model Matrices A & B Specified	**VERIFIED**	Full Multimodal (26 feats) vs Pure Non-ST (20 feats)
Machine Learning Models Trained	**ZERO**	Halted awaiting explicit human authorization
BeatAhead Website Modified	**ZERO**	Web application remains completely untouched

Recommended Next Action

Present this split audit for human review. Once authorized, proceed to Phase 5 modeling using **5-fold StratifiedGroupKFold cross-validation on an 80-case development cohort plus a locked 20-case test set**, benchmarking Model Matrix A against Model Matrix B.